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Ever Diligent Consulting, LLC
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<https://RAGaaS.app>



Executive Summary:

This document is a response from Ever Diligent Consulting, LLC to a Request for Information (RFI) on an AI Action Plan. It focuses on integrating Retrieval Augmented Generative (RAG) models using Llama 3.2:latest and building an OCR application using the Llama 3.2-Vision model from Meta. The document concludes by providing actual output delivered by Ever Diligent Consulting, LLC. Flagship Retrieval Augmented Generative (RAG) application “RAGaaS.app” and emphasizing the transformative opportunities of these technologies and encouraging further exploration and implementation.

Key points include:

- **AI Research and Development:** Utilizing RAG models for accurate document retrieval and OCR applications for text recognition and image analysis.
- **Workforce Development and Training:** Emphasizing training programs for RAG and OCR technologies to enhance efficiency and accuracy.
- **AI Governance and Ethics:** Ensuring ethical use with validation mechanisms and privacy protocols.
- **International Collaboration:** Promoting partnerships and open-source initiatives.
- **National Security:** Applying RAG for real-time access to classified documents and OCR for situational awareness.
- **AI Applications Across Sectors:** Highlighting applications in healthcare and transportation.
- **Data Access and Utilization:** Ensuring secure and high-quality data access.
- **Public Engagement and Awareness:** Educating the public about RAG and OCR technologies.
- **AI-Driven Innovation Ecosystem:** Fostering innovation through sandboxes and collaborations.

- **AI and National Security:** Enhancing intelligence analysis and surveillance through RAG and OCR.

Introduction:

Thank you for the opportunity to contribute to the development of an AI Action Plan. This response emphasizes two key areas: integrating Retrieval Augmented Generative (RAG) models using Llama 3.2:latest and building a powerful OCR application using the Llama 3.2-Vision model from Meta. These technologies hold significant potential for enhancing various sectors, including defense, education, governance, and national security.

1. AI Research and Development:

- **Integration of RAG Models:**
 - Ever Diligent Consulting's innovative use of Llama3.2:latest, GPT-4 and Ada-002 embeddings exemplifies how RAG models can revolutionize document retrieval. By connecting LLMs to vast repositories, this system ensures factual accuracy and prevents hallucinations.
 - RAG models combine generative AI with retrieval techniques, enabling systems to access external knowledge bases for accurate, context-aware responses. For defense applications, this integration ensures real-time access to classified documents, enhancing decision-making processes. By validating outputs against authoritative sources, RAG systems maintain transparency and reliability, critical for national security.
 - **Benefits include:**
 - Precision: With 90%+ search accuracy, defense personnel can quickly access relevant technical documents.
 - Efficiency: Instant query mapping enables rapid retrieval of equipment specs and battlefield reports.
 - Insight Extraction: RAG technology aids in analyzing complex data, providing valuable insights from various reports.
- **OCR Application with Llama 3.2-Vision:**
 - The Llama 3.2-Vision model enables advanced OCR capabilities, allowing text recognition and image analysis directly on personal devices. This tool supports efficient extraction of actionable intelligence from visual data, enhancing situational awareness in critical scenarios. Its versatility makes it suitable for a wide range of applications, including document processing and object detection.
 - **Advantages include:**
 - Efficiency: Developed locally, this tool leverages advanced AI for tasks that were once complex.

- **Versatility:** Experimentation with different images and prompts allows full exploration of the model's potential.

2. Workforce Development and Training:

- **RAG Integration in Training:** Training programs should focus on developing skills for RAG systems and OCR applications. Personnel must be proficient in leveraging these tools to enhance efficiency and accuracy in defense operations. Tailored curricula can equip professionals with the necessary expertise to handle complex data extraction tasks and interpret generative AI outputs effectively.
 - Incorporate RAG models into training programs to provide defense personnel with accurate, context-specific responses.
 - This approach enhances learning by offering clear explanations derived from authoritative documents.
- **OCR Application Education:**
 - Develop a curriculum focused on building and utilizing OCR applications using Llama 3.2-Vision.
 - Empower professionals to handle visual data effectively, enhancing their skills in extracting information from images.

3. AI Governance and Ethics:

- **RAG Validation:** Ensuring ethical use of RAG and OCR technologies requires robust validation mechanisms and strict privacy protocols. For RAG systems, transparency in document retrieval processes is essential to maintain trust. In OCR applications, safeguarding sensitive information extracted from images must be prioritized to prevent data breaches.
 - Ensure RAG systems validate outputs against authoritative documents to maintain transparency and accountability.
 - This practice is crucial for maintaining trust and reliability in defense applications.
- **OCR Privacy and Security:**
 - Emphasize the importance of privacy and security when extracting sensitive information from images.
 - Develop guidelines to protect data integrity during OCR processes.

4. International Collaboration Between Government and Industry:

- **RAG Partnerships:** Partnerships between government agencies and private sector entities can accelerate advancements in RAG and OCR technologies. Collaborative efforts can lead to the development of tailored solutions that meet specific defense needs while fostering innovation in AI research. Open-source initiatives around OCR tools, for example, can encourage shared development and best practices among stakeholders.
 - Encourage partnerships to develop RAG systems tailored to defense needs, ensuring they meet high standards for accuracy and reliability.
- **OCR Open-Source Projects:**
 - Promote open-source initiatives around OCR tools using models like Llama 3.2-Vision.
 - Foster innovation by sharing resources and best practices among stakeholders.

5. Addressing National Security Challenges:

- **RAG in Defense:**
 - Utilize RAG systems for real-time access to classified documents, enhancing decision-making processes.
 - This capability can provide a strategic advantage by ensuring timely and accurate information retrieval.
- **OCR in Situational Awareness:**
 - Deploy OCR applications to enhance situational awareness by quickly extracting actionable intelligence from visual data.
 - This technology supports rapid response and informed decision-making in critical scenarios.

6. AI Applications Across Sectors:

In various sectors, integrating Retrieval-Augmented Generative (RAG) models using Meta's Llama 3.2 and Optical Character Recognition (OCR) applications with the Llama 3.2-Vision model can drive significant advancements.

- **Healthcare:** RAG can enhance medical diagnostics by accessing a comprehensive knowledge base of symptoms and treatments, enabling more accurate diagnoses. OCR can facilitate digitizing old medical records, making historical data accessible for analysis.
- **Transportation:** RAG can optimize logistics by retrieving real-time traffic data to suggest efficient routes. OCR can assist in reading license plates or signs, improving safety and operational efficiency.

7. Data Access and Utilization:

Ensuring high-quality and secure data access is crucial for both RAG and OCR technologies:

- RAG: The integration of Llama 3.2 with external datasets requires robust data pipelines to maintain accuracy and relevance. Implementing strict privacy protocols ensures sensitive information remains protected.
- OCR: High-quality images are essential for accurate text extraction. Anonymizing data before processing can address privacy concerns, especially when dealing with sensitive documents or records.

8. Public Engagement and Awareness

Educating the public about RAG and OCR technologies is vital for their successful adoption:

- Workshops and Resources: Hosting workshops and creating online resources that explain how RAG and OCR work in simple terms can demystify these technologies.
- Addressing Concerns: Discussing potential benefits alongside privacy and ethical considerations can alleviate concerns, fostering trust in AI applications.

9. AI-Driven Innovation Ecosystem

Fostering an environment conducive to innovation is key to advancing RAG and OCR technologies:

- Sandboxes and Test Environments: Providing spaces for developers to experiment with Llama 3.2 and OCR models can accelerate innovation.
- Collaborations: Encouraging partnerships between academia, government, and private companies can drive advancements and practical applications of these technologies.

10. AI and National Security

Integrating RAG and OCR into national security frameworks offers enhanced capabilities:

Intelligence Analysis: Using RAG to analyze intelligence data by accessing historical information can improve decision-making processes.

Surveillance and Document Processing: OCR can efficiently process large volumes of documents or images, aiding in surveillance efforts. Ensuring ethical use and proper oversight is essential for maintaining public trust.

RAG Outputs: Please zoom in to see how the files the LLMs query against are to the right in the screenshot below.

The screenshot displays a RAG application interface. On the left, a sidebar titled "Data Groups" lists various document types: Federally Funded..., Contract Review..., Article and Clari..., Artificial Intellig..., Image creator..., OCR Scanner..., and Cart. The main content area shows a document titled "Federally Funded Research Data Centers (FFRDCs)". The document text includes:

Key Characteristics of FFRDCs:

- Collaborative:** FFRDCs often collaborate with government agencies, universities, and private industry partners to conduct research that addresses significant national challenges.
- Data Access:** These centers typically provide secure access to sensitive data, allowing researchers to work on projects that require confidentiality or involve classified information.
- Expertise and Talent:** FFRDCs employ highly skilled scientists, engineers, and researchers who possess specialized knowledge in various fields, including science, technology, engineering, and mathematics (STEM).
- Funding:** They receive funding from federal agencies, which helps to sustain their operations and research activities. The funding is often tied to specific projects or areas of interest to the government.
- Public Benefit:** The research conducted at FFRDCs is intended to advance knowledge and technology for the benefit of the public, often leading to innovations that improve national security, public health, and the economy.

Examples:

- MIT Lincoln Laboratory:** Focuses on advanced technology and defense projects.
- Admiral Commission:** Engages in policy research and advises senior civilian leaders.

FFRDCs play a vital role in advancing the scientific ambitions and security objectives of the United States, acting as a bridge between academia and government needs.

Specific Members:

The specific members serving on these committees may vary from session to session, and leadership positions such as the committee chair and steering committee will have a significant role in overseeing FFRDCs. To get the most accurate and up-to-date information, it's recommended to check the official U.S. House of Representatives or the respective committee websites, as memberships and committee assignments can change frequently.

Key Roles of the Defense Technical Information Center (DTIC):

- Information Repository:** DTIC serves as a central repository for defense-related research and technical information. It collects, organizes, and disseminates information from various sources, including government agencies, universities, and private industry.

On the right side of the interface, a "Setup Detail" panel shows a list of files used for retrieval:

- What are Federally Funded Research Data Centers (FFRDCs)?
- What does the defense technical information center (DTIC) do?
- What does the defense technical information center (DTIC) do?

Below the list, there is a "Feedback" button and a "Next" button.

OCR Prototype Outputs:

Output 1

Image Quality

For the vision impaired the image may come across as light/blurry

Accuracy Assessment /Score

Good @ subjective 92%

Solution to Make It Better:

1. Scan image with industry leading scanner.
2. Further Train the LLM with similar pictures.

The image displays a handwritten medical prescription form and its corresponding OCR output. The form is titled 'Rx' and contains the following information:

- Name: Armando Ceguia
- Address: Unit Rimbo, Makati City
- Age: 29 Sex: M Date: 12-03-90
- Rx: (Trimox)
- Amoxicillin 500mg Cap #21
- Sig: 1 cap 3x a day for seven days.
- Physician's Sig: [Signature]
- Lic No: 1234567
- PTR No: 1234567
- S2 No: [blank]

The OCR output on the right is a structured text representation of the form's content, showing the model's response for each field. The output is as follows:

```

(portfolio) ..... MacBook-Pro llama-ocr % python llama-ocr.py
Model Response: Name: Armando Ceguia
Address: Unit Rimbo, Makati City
Age: 29
Sex: M
Date: 12-03-90

Rx:

(Trimox)
Amoxicillin 500mg Cap #21
Sig: 1 cap 3x a day for 7 days.

Physician's Signature: [unclear]

License Number: 12345
PTR No.: 1234567
S2 No.: [blank]
(portfolio) ..... MacBook-Pro llama-ocr %
  
```


Output 2

Image Quality

For the vision impaired the image may come across as light/blurry

Accuracy Assessment / Score

Very good @ subjective 95%

Solution to Make It Better:

1. Scan image with industry leading scanner.
2. Further Train the LLM with similar pictures.

Reynaldo O. Joson, M.D.
MANILA DOCTORS HOSPITAL
 Suite 301, Medical Arts Center, U.N. Avenue Ermita, Manila
 Telephone # 524-30-11 Loc 4100; 522-47-13

Ma. Li... a Ma'n Date 12-6-2014
 561F
 Concern: Neck mass - noted 2 yrs ago
 no pain
 Physical Exam Findings (Neck):

Diagram: A hand-drawn diagram of a neck cross-section. A circle labeled 'R' is on the left, and a circle labeled 'L' is on the right. An arrow points from the 'R' circle to a box labeled 'Thyroid Mass Right Lobe'. Below the 'R' circle, it says 'no tenderness' and 'no redness'. Below the 'L' circle, it says 'CANCER' and 'NOT CANCER'. A line connects the two circles, with 'CANCER' written above it and 'Need for atest!' written below it.

Options for further Diagnostic Test:
 Biopsy, Risk, Lab Tests, Pain, T tests, Bladder, Infusion, IT, fresh puncture, Thyroid Disorder, Function, Hormone, Hypo < No symptoms = For you=Normal, Menses

Thyroid Disorder Function Hormone Hypo < No symptoms = For you=Normal Menses

Lic. # 44609

Model Response: Reynaldo O. Joson, M.D.
 MANILA DOCTORS HOSPITAL
 Suite 301, Medical Arts Center, U.N. Avenue Ermita, Manila
 Telephone # 524-30-11 Loc 4100; 522-47-13
 Ma.Li... a Ma'n
 Date: 12-6-2014
 14
 Concerns:
 Neck mass-noted 2 yrs ago
 no pain
 Physical Exam Findings (Neck):
 Thyroid Mass Right Lobe
 Cancer
 Need for atest!
 Optime for further diagnostic Test:
 Biopsy
 Risk
 Pain
 Bladder
 Infusion
 IT
 fresh puncture
 Thyroid Disorder
 Function
 Hormone
 Hypo < No symptoms = For you=Normal
 Menses
 Lic. # 44609

Output 3

Image Quality

For the vision impaired the image may come across as light/blurry and very fuzzy

Accuracy Assessment /Score

Below Average @ subjective
79%

Solution to Make It Better:

1. Scan image with industry leading scanner.
2. Further Train the LLM with similar pictures.

```

llama-ocr --zsh - 79x48
(portfolio) phuqoffcowards@Charless-MacBook-Pro llama-ocr % python llama-ocr.py
Model Response: NORMAN D. BABAR, M.D., Dpps
PEDIATRICIAN
CLINIC ADDRESS: Suite 204 Medical Arcade
J. Carreto Avenue, General Santos City Tel No.: (083) 553-1858
CLINIC HOURS:
Monday to Saturday 10:30 AM - 12:00 NN
2:30 PM - 5:00 PM
AFFILIATIONS:
St. Elizabeth Hospital
Soccksargen Hospital
MMC Baptist Hospital
GenSan Doctor's Hospital
Diagan Hospital
Dr. Jorge Royeca Hospital
Patient's Name:
Date:
Address:
Age:
Sex:
Rx
Next Visit:
NORMAN D. BABAR, M.D.
Lic # 93566
PTR No. 9559583
32 No.

```

Output 4

Image Quality

Blurry

Accuracy Assessment /Score

Almost Above Average @ subjective 87% as the man does not have a red shirt.

Solution to Make It Better:

1. Take image with industry leading camera.
2. Further Train the LLM



Model Response: The image depicts a man sitting at a desk, working on his computer.

The man is wearing a blue suit jacket over a white collared shirt and has short black hair. He is seated in a gray office chair with his hands resting on the keyboard of an open laptop that sits on top of a white desk. The laptop's screen displays a colorful graphic design. To the right of the laptop, there is a large monitor displaying a similar design.

The background wall behind the man features a light-colored geometric pattern resembling circuit boards or wires.

[]:

Output 5

Image Quality

Kind of Crisp

Accuracy Assessment /Score

Excellent @ subjective 100%

Solution to Make It Better:

1. Take image with industry leading camera.
2. Further Train the LLM



```
50-oct
s0Charles-MacBook-Pro llama-ocr % ls
12.51.33 AM.png
12.52.58 AM.png
12.55.05 AM.png
```

Model Response: Google Introducing Quantum AI Chip - Willow

* ****Image Description:****

* The image features a large screen displaying an advertisement for Google's new Quantum AI chip, Willow.

* On the left side of the screen is a 3D rendering of the chip itself, which appears to be a green cube with a black square in the center. Inside this square is the Google logo, comprising four colored semicircles (red, yellow, blue, and green) arranged in a circular pattern.

* To the right of the chip is the title "Google Introducing Quantum AI Chip - Willow" in large text. The word "Quantum" is highlighted in neon green, while the rest of the text is white.

****Transcription:****

Google
Introducing
Quantum
AI Chip
- Willow

Simple Python 3 (invkernel) Idle

Mode: Edit Ln 1 C

Output 5

Image Quality

Kind of Crisp

Accuracy Assessment /Score

Excellent @ subjective 100%

Solution to Make It Better:

Take image with industry leading camera.

Further Train the LLM

Image Example: Somewhat Complex Technical Report

Article

<https://doi.org/10.1038/s41467-025-56811-z>

using PMMA as the e-beam resist followed by e-beam evaporation of Au with a 2 nm Ti adhesion layer. The thicknesses t_1 for the gate dielectric separating the gate electrode (also underneath mirror) and graphene layer and t_2 for the separation between the graphene layer and the Au metasurface are summarized for each device: ($t_1 = 20$ nm, $t_2 = 10$ nm) for Figs. 1b, 6, 7 and Supplementary Figs. 2, 6, 7; ($t_1 = 20$ nm, $t_2 = 0$ nm) for Fig. 2; ($t_1 = 150$ nm, $t_2 = 50$ nm) for Figs. 3 and 4; ($t_1 = 100$ nm, $t_2 = 10$ nm) for Fig. 5; ($t_1 = 40$ nm, $t_2 = 10$ nm) for Fig. 8 and Supplementary Fig. 8; ($t_1 = 100$ nm, $t_2 = 0$ nm) for Supplementary Figs. 3–5. Finally, the source and drain electrodes are patterned using the same method as the gate electrodes.

Thermal and optical simulation

The transient electrothermal simulation is conducted via time-domain finite-element analysis in COMSOL Multiphysics. The single pixel Gr-FET device in Fig. 7 is reconstructed in a COMSOL model, with the constant temperature boundary condition applied to the bottom surface of the simulated device. The electrical properties and setups of graphene applied in simulation are obtained from experimental designs and measurements in Supplementary Information. Material properties of Si, SiO₂, Au and Al₂O₃ are obtained from COMSOL material database.

The optical simulation is based on the Ansys Lumerical FDTD. A single period of the MIM metasurface is illuminated by a plane wave source with periodic boundary conditions. A reflective power monitor is placed above the source to capture the reflective energy, which provides information about the emissivity based on Kirchhoff's law, revealing coherent features of the thermal infrared emission. For angle-dependent simulations, we change the periodic boundary conditions to Bloch boundary conditions to properly account for the phase correction.

Spectroscopy and thermal mapping measurements

All the spectra are measured by the Thermo Fisher Nicolet iSSO Fourier transform infrared spectrometer. A microscope with an adjustable aperture focuses infrared illumination on the active area of the MIM metasurface. The reflectance R is measured under different polarization angles based on a polarization filter. By the principle of energy conservation, we can obtain the absorptance as $1 - R$. Since all devices are reciprocal and in thermal equilibrium, Kirchhoff's law, which states that emissivity equals absorptance, can be applied. The current-voltage sweeps are measured by the Keysight A1500A semiconductor device analyzer. The custom-designed PCB testing system consists of software codes and control circuits. During testing, control commands (codes) are first compiled by Arduino software and then sent to an ATmega-2560 micro-controller unit (MCU) development board. For the gate control, 16 bits digital signals/data are sent to 3 digital-to-analog converters (DACs) that are simply made by digital potential meters, respectively, from the MCU. Then, 3 OPAMP amplifiers map the 0 ~ 5 V analog voltages from the DACs to ~5 ~ 5 V gate control. For graphene device source and drain controls, each of its source pads is connected to an N-MOSFET. By wiring an output port of the MCU to the gate of the N-MOSFET, each source and drain can be turned on or off individually. The static and dynamic thermal mapping images are then captured by the QFI InfraScope system. In the time-domain thermoreflectance measurements, a 530-nm monochrome incident LED lights up the pixel area before being reflected and captured by a CCD camera. The reflectivity change is linearly proportional to the surface temperature change: $\Delta R/R = C_{TR} \Delta T$, where C_{TR} is the thermoreflectance coefficient depending on surface material and beam wavelength. Au has a local minimum of negative C_{TR} under 530-nm incident wavelength, hence $\Delta R/R$ will decrease with increase of temperature change, and vis versa. All of the measurements are taken at room temperature.

Data availability

The Source Data underlying the figures of this study are available with the paper. All raw data generated during the current study are available from the corresponding authors upon request. Source data are provided with this paper.

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```

cleaned_text = response['response'] + LLM_PROMPT_SUFFIX()
return {'model': 'gpt-4o', 'cleaned_text': cleaned_text}

Model Response: **Transcription:**

Article
using PMMA as the e-beam resist followed by e-beam evaporation of Au with a 2nm Ti adhesion layer. The thickness t1 for the gate dielectric separating the gate electrode (also underneath mirror) and graphene layer and the Au metasurface are summarized for each device:

(t1 = 20 nm, t2 = 10 nm) for Figs. 1b, 6, 7 and Supplementary Figs. 2, 6, 7;

(t1 = 20 nm, t2 = 0 nm) for Fig. 2; (t1 = 150 nm, t2 = 50 nm) for Figs. 3 and 4; (t1 = 100 nm, t2 = 10 nm) for Fig. 5; (t1 = 40 nm, t2 = 10 nm) for Supplementary Figs. 8 and Supplementary Figs. 3-5. Finally, the source and drain electrodes are patterned using the same method as the gate electrodes.

Thermal and optical simulation
The transient electrothermal simulation is conducted via time-domain finite-element analysis in COMSOL Multiphysics. The single pixel GrFET device in Fig. 7 is reconstructed in a COMSOL model, with the constant temperature boundary condition applied to the bottom surface of the simulated device. The electrical properties and setups of graphene applied in simulation are obtained from experimental designs and measurements in Supplementary Information. Material properties of Si, SiO2, Au and Al2O3 are obtained from COMSOL material database.

The optical simulation is based on the Ansys Lumerical FDTD. A single period of the MIM metasurface is illuminated by a plane wave source with periodic boundary conditions. A reflective power monitor above the structure captures the reflected energy, which provides information about the emissivity based on Kirchhoff's law, revealing coherent features of thermal infrared emission. For angle-dependent simulations, we change the periodic boundary conditions to properly account for phase correction.

Spectroscopy and Thermal Mapping Measurements
All spectra are measured by the Thermo Fisher Nicolet 1550 Fourier transform infrared spectrometer. A microscope with an adjustable aperture focuses infrared illumination on the active area of the MIM metasurface. The reflectance R is measured under different polarization angles based on a polarization filter. By the principle of energy conservation, we can obtain the absorbance as 1-R. Since all devices are reciprocal and in thermal equilibrium, Kirchhoff's law states that emissivity equals absorbance, which can be applied.

Data Availability
The source data generated during this study is available from the corresponding authors upon request.

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Conclusion:

Finally, the integration of RAG models and the development of powerful OCR applications represent transformative opportunities for various sectors. By investing in these technologies, the U.S. can maintain its leadership in AI while addressing critical challenges in defense, education, governance, and national security. We encourage further exploration and implementation of these innovations to unlock their full potential.